# CMPSC 421 – Netcentric Programming

## Homework 2

The following problem set is worth 100 points. The assignment involves querying data for applications using AJAX techniques.

### AJAX – Asynchronous JavaScript and XML

Most modern web applications today query data on the client side using AJAX from an API. This assignment involves you producing four different applications using a variety of APIs.

### 1. Weather Application

Several smart home devices (such as Amazon Alexa and Google Home) have useful interfaces for viewing weather data. For instance, the *Amazon Echo Spot* displays time and temperature using this screen:

Check time, weather,
and more at a glance

And the *Amazon Echo Show* displays a more detailed weather forecast with an interface like this:

To retrieve this weather data, you'll need to query an external API. One useful API that is available is the **Open Weather Map API**. An example of an API call would be the following:

> *https://api.openweathermap.org/data/3.0/onecall?lat=33.44&lon=-94.04&exclude=hourly,daily&appid={API key}*

This returns the current weather data given latitude and longitude. Of course you'll need to receive that from the user, and its much more intuitive to prompt the user for a zip code instead.

### *Assignment Details*

For the first assignment, produce a web application that does the following:

- Ask the user for a zip code
- Using an external API (e.g. Mapquest), query the user's lat/lon using the zipcode
- With the lat/lon query the weather forecast
- Display the forecast in a UI structure of your choosing

## 2. Dog API

The next API you will use is an API that display **breeds of dogs**. As an example:

> *https://dog.ceo/api/breeds/list/all*

This returns a JSON data set of all dog breeds. You can then display any images of the breed using this API call:

> https://dog.ceo/api/breed/hound/images

For this assignment:

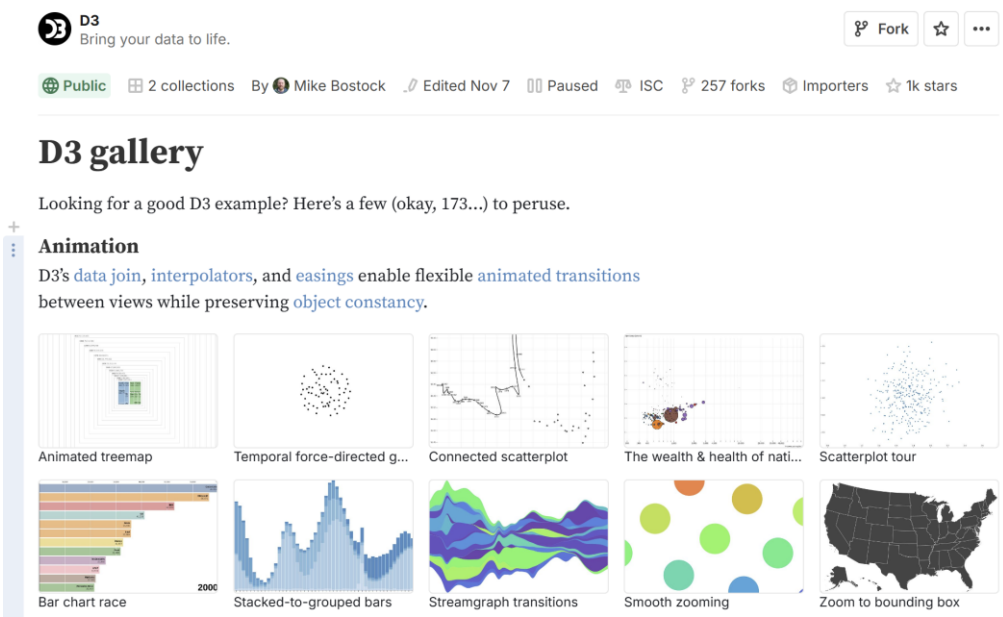
- When the page loads, query a list of dog breeds and display their images in a grid using Materialize CSS cards
- Add a **filter box** on top of the grid that dynamically filters images based on the breed the user types
- Feel free to use other APIs to obtain more information about the dog itself, since the current API only provides breed and images

## 3. Country Data with D3.js

An API is available to retrieve information about all the countries in the world, such as their cities, capitals, languages, currencies, etc.

https://restcountries.com/#rest-countries

As mentioned in class, a JavaScript framework exists for displaying data using **D3.js**. This framework is widely used on several applications including **The New York Times**.




One module in D3 you can use is the **d3-geo** library. This library can be used for producing interactive maps:

https://d3js.org/d3-geo

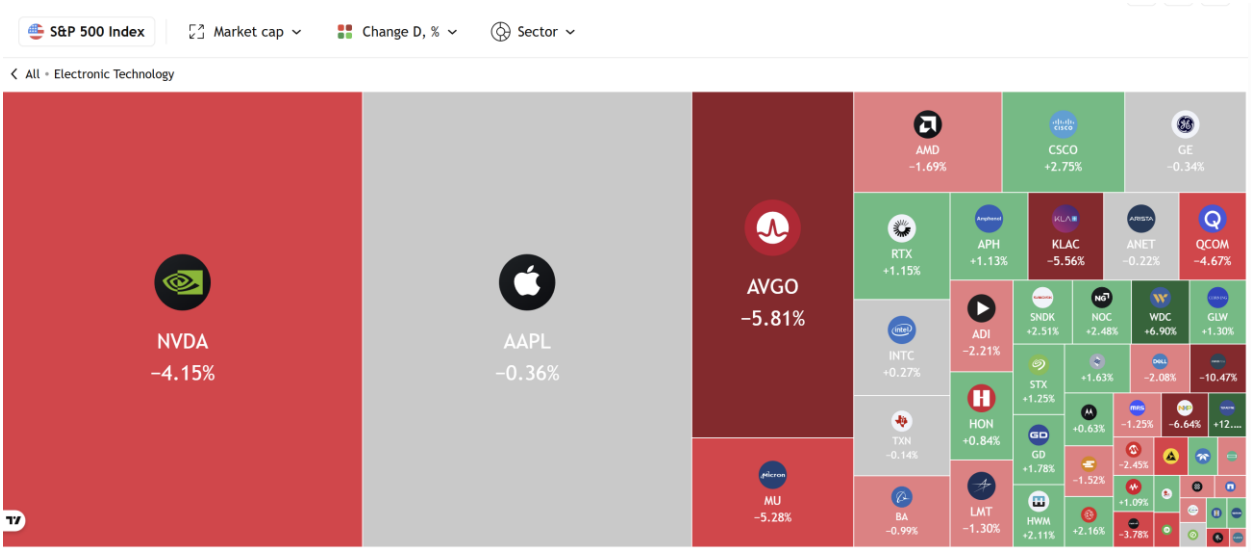
*Assignment Details*

For this assignment:

- Produce an interactive map of the globe with all countries (using D3)
- When the user interacts with a country, query for all information about the country (such as flags, languages, etc)
- Display the detailed information about the country that you receive from the countries API

## 4. Crypto Heat Map

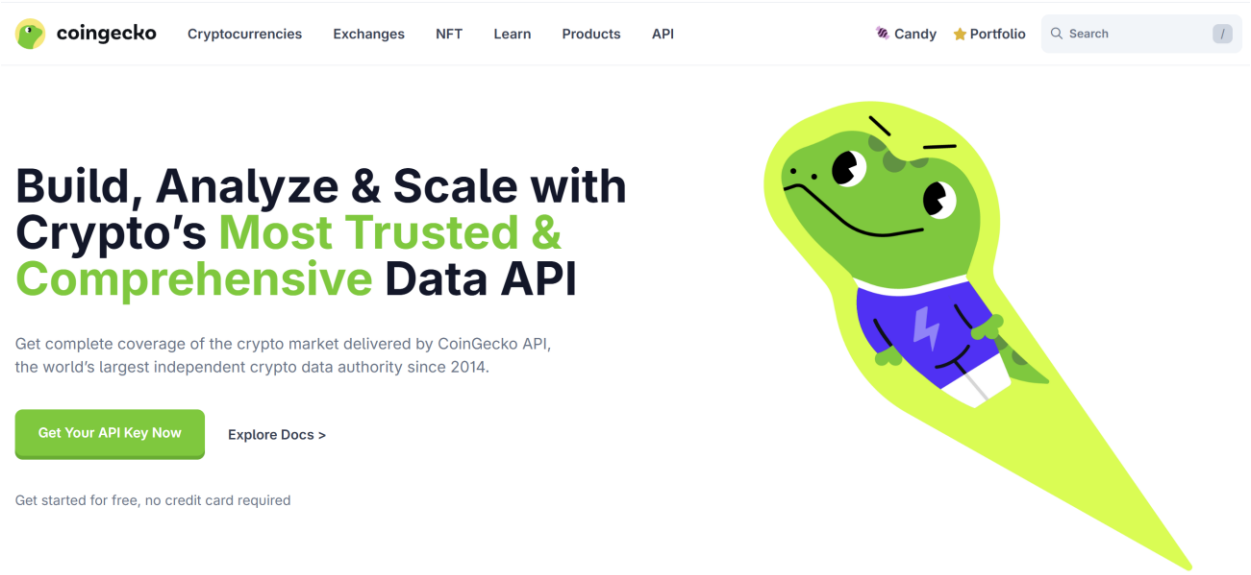
In trading, a **heatmap** is an interactive graph used to show daily activity across a major index, such as (in this case) the S&P 500:




This map is generated dynamically using:

- The volume (larger volume = larger box)
- Daily Change (positive change from previous day = green, red if otherwise)

For this assignment, you are to make the same box for **crypto currencies**. One particular API you can use to retrieve data for crypto is **Coin Gecko**:
https://www.coingecko.com/en/api




For this assignment, simply produce a daily heat map for the major crypto currencies that you receive from the API. The larger the box, the more prominent the crypto. For instance, the largest box on your heat map will be Bitcoin (BTC). The second-largest box is Ethereum (ETH), since these are the two largest cryptocurrencies by volume.

## Assignment Details

Upload a zip file of all source code I need to run your four applications before the close of the Canvas dropbox.